

From Roles and Realizers to Electrochemical Brain Waves

A Four-Lens Comparison of Ned Block's 2025-2026 Consciousness Argument and Self-Aware Networks (2011-2025)

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

Ned Block's *Can only meat machines be conscious?*, first published online in October 2025 and assigned to the April 2026 issue of *Trends in Cognitive Sciences*, argues that computational roles may not suffice for consciousness and that subcomputational biological realizers may be necessary [5]. A 2026 follow-up makes the mechanism more explicit: electrochemical processing may be required for brain rhythms, brain rhythms may be required for sophisticated cortical processing, and those mechanisms may be necessary for consciousness [6]. This paper compares that development with Micah Blumberg's dated Self-Aware Networks (SAN) record. The comparison uses four distinct lenses: operational mechanism, causal-interface narrative, conclusion-first forensic reconstruction, and longitudinal signature migration. The result is bounded but significant. Block's anti-functionalism and phenomenal / access distinction long predate SAN, and his 1995 discussion properly attributes oscillatory binding to Crick and Koch [1,2,7]. Those broad atoms are not claimed for Blumberg. However, Blumberg's public April 2017 Neural Lace Podcast already joined a time-varying three-dimensional ionic-charge pattern, electromagnetic or brain-wave communication, feedback, and a conscious-versus-unconscious protocol question [14]. Immutable 2022 Git records then added phase-wave differentials, distributed receive-transform-project operations, and Neural Array Projection Oscillation Tomography [15,16]; the 2024 book consolidated a molecular-to-oscillatory bridge [17]. Block's 2025-2026 work later converges on a bounded portion of this joined causal chain: ionic or electrochemical neural state, population waves or rhythms, coordinated processing, and consciousness relevance [5,6]. The target papers do not instantiate the full NAPOT / PWD operator or Blumberg's potassium-to-action-potential-duration-to-calcium route. The supported conclusion is therefore a Micah-earlier partial priority on a specific causal subchain, not identity of the complete theories and not a claim about access or intent. Because the candidate atoms were selected after inspection and lack empirical base-rate and dependency calibration, no numerical probability of independent derivation is reported.

Probability status: quantitative-not-calibrated.

Keywords: consciousness; biological computation; electrochemical processing; brain rhythms; neural oscillations; Self-Aware Networks; NAPOT; phase-wave differentials; priority; conceptual genealogy

Scope and evidence boundary

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1. Introduction

Comparisons between theories of consciousness often fail in one of two directions. A comparison may be too coarse, treating every reference to biology, oscillation, or information as the same idea. Or it may be too strict, demanding that a later author reproduce an earlier theory's complete vocabulary and formalism

before any relationship is admitted. The first error manufactures similarity. The second hides real partial equivalence behind surface differences.

This paper uses an atomized middle course. It asks whether a dated earlier source and a later target share an input-transform-output subchain, a causal narrative, or a conclusion-level functional relationship. It then records the residuals rather than allowing them either to erase the match or to be silently imported into the target.

The focal target is Ned Block's 2025-2026 argument that consciousness may depend on subcomputational biological mechanisms [5,6]. The strongest target-side passage is not merely the familiar claim that substrate matters. The 2026 follow-up states a causal sequence: sophisticated cortical processing requires brain rhythms; brain rhythms depend on electrochemical processing; and consciousness may depend on those mechanisms [6]. That sequence is compared with the SAN genealogy, especially Blumberg's public April 2017 description of a changing ionic-charge matrix linked to brain-wave / electromagnetic communication and conscious feedback [14], followed by the 2022 NAPOT / PWD formulation [15,16] and the 2024 molecular consolidation [17].

The paper makes four distinctions throughout:

1. priority on a broad field atom versus priority on a joined, higher-specificity architecture;
2. partial equivalence versus identity of complete theories;
3. content and chronology versus access or intent; and
4. a transparent qualitative evidential judgment versus an uncalibrated numerical probability.

2. Source record and chronology

2.1 Block's earlier work

Block's earlier record prevents an exaggerated conclusion. *Troubles with Functionalism* (1978) argued that functional characterizations can be too liberal or too chauvinistic [1]. *On a Confusion about a Function of Consciousness* (1995) distinguished phenomenal consciousness from access consciousness [2]. It also discussed Crick and Koch's proposal that synchronized activity in the 35-75 Hz range might address feature binding [2,7]. Block did not present that oscillation proposal as his own mechanism. He explicitly asked why one frequency rather than another should underlie experience and entertained the possibility of physiologically similar brains with different experience [2].

In *Consciousness, Accessibility, and the Mesh between Psychology and Neuroscience* (2007), Block sought neural realizers of phenomenal consciousness distinct from mechanisms of cognitive accessibility [3]. Forty-hertz activity appeared as a possible neural-correlate item, not as an ionic-wave theory [3]. In the 2018 coauthored letter *Conscious Machines: Defining Questions*, Block and colleagues argued that machine-consciousness research must ask whether a system has subjective experience; global availability and metacognition were neither necessary nor sufficient [4]. The letter did not specify electrochemical processing, extracellular ion waves, or brain rhythms as the required mechanism [4].

The pre-2025 record therefore contains anti-functionalism, phenomenal-consciousness analysis, neural-realizer methodology, and awareness of oscillatory NCC proposals. It does not yet contain Block's later electrochemical-to-brain-wave causal chain.

2.2 The 2025-2026 Block mechanism

Can only meat machines be conscious? distinguishes computational roles from subcomputational biological realizers [5]. Block calls his proposed orientation "subcomputationalism" and suggests that fluctuations

in the ionic composition of extracellular fluid could be important to consciousness, while denying that the particular ions used by extant organisms must be necessary [5]. He describes electrochemical neural signaling, asks what made it evolutionarily advantageous, and connects chemical synapses to brain rhythms. He credits R. Douglas Fields for the claim that ionic fluctuations in extracellular fluid are important to population activity and credits Peter Godfrey-Smith for a consciousness-relevant role for large-scale dynamic patterns [5,9-11]. The paper also preserves a contrary-data boundary by discussing Kiper and colleagues’ negative result for flicker-driven phase coherence in visual grouping [5,8].

The 2026 follow-up is more direct. It proposes that ionic fluctuations, glial regulation, and subthreshold oscillations contribute to population-wide electrical waves that help drive neural firing [6]. Its explicit argumentative chain can be represented as:

$$E \longrightarrow R \longrightarrow S \longrightarrow C?, \quad (1)$$

where E is electrochemical processing, R is brain-rhythm generation, S is sophisticated cortical processing, and $C?$ is the openly speculative claim that consciousness depends on those mechanisms. The question mark is essential: Block labels the final steps speculation.

2.3 The SAN genealogy

SAN is not a single 2025 snapshot. Its relevant genealogy develops in stages.

- **2011:** a predictive-self and whole-process observer ancestor appears in *The Self Is a Prediction*. This is conceptual ancestry, not yet the later ionic-wave mechanism [12].
- **2012:** Neo Mind Cycle videos describe recursive visual-motor feedback, EEG-driven light and sound as a two-way interaction, coordinated neural electromagnetic waves, and an internally projected experienced world [13].
- **April 11, 2017:** Neural Lace Podcast 1 describes a three-dimensional neural matrix in which each point carries a changing time-and-space pattern of ionic charges. The same episode asks what brain waves communicate with, treats electromagnetic-frequency communication as a protocol problem, and contrasts feedback-connected conscious communication with a less connected unconscious mode [14].
- **August-November 2022:** immutable Git sources define phase-wave differentials as content-bearing differences, describe dendrites and arrays as receivers and transformers, describe terminals and downstream arrays as projectors, and name NAPOT as a distributed rendering architecture without a central viewer [15,16].
- **2024:** *Bridging Molecular Mechanisms and Neural Oscillatory Dynamics* consolidates the molecular-to-wave program, including ion-channel and action-potential dynamics, synaptic release, plasticity, phase relations, and multiscale oscillatory organization [17].
- **2025:** SAN / OCA drafts consolidate multiscale agency, embodied feedback, and prediction-error reduction [18].

The earliest public source needed for the present bounded comparison is the 2017 podcast. The 2022 and 2024 sources increase specificity; they do not replace the earlier partial priority.

3. Four-lens comparison method

3.1 Claim atoms and partial equivalence

Represent a claim atom as an ordered causal triple:

$$a = (I, T, O), \quad (2)$$

where I is an input or physical condition, T is a transformation, and O is an output or functional consequence. Two complete theories need not be identical for a bounded subchain to match. For a Micah atom a_M and target atom a_B , a partial operational match exists when a nonempty ordered subchain is preserved:

$$P(a_M, a_B) = 1 \quad \text{iff} \quad \exists s \neq \emptyset : s \preceq a_M \wedge s \preceq a_B. \quad (3)$$

Here $\preceq$ means that the same causal ordering and functional role are retained after terminology is translated. Equation (3) does not allow details present only in SAN to be attributed to Block.

3.2 Three content lenses

The content determination uses three co-equal lenses:

- **Mechanism lens:** does the target preserve the same input-transform-output behavior?
- **Narrative lens:** does the same causal-interface architecture connect the components even when ontology or vocabulary differs?
- **Detective lens:** starting from a shared conclusion, does working backward recover a bounded earlier chain and distinguish true residuals from surface redescription?

A bounded content match is retained if any one lens passes:

$$M = L_{\text{mechanism}} \vee L_{\text{narrative}} \vee L_{\text{detective}}. \quad (4)$$

Passing more than one lens increases explanatory robustness but is not converted into a probability.

3.3 Longitudinal fourth lens

The fourth lens is program-level and chronological. It compares a target author's earlier and later stages around Micah's dated record. Let A_t be the set of affirmed mechanism atoms at target stage t . A component shift toward a Micah cluster K_M occurs when:

$$|A_{\text{after}} \cap K_M| > |A_{\text{before}} \cap K_M|, \quad (5)$$

provided the stage dates and source texts are independently documented. Equation (5) records a directional change. It does not establish access, copying, or intent.

3.4 Priority predicate

For a matched bounded atom s , Micah-earlier priority is recorded when:

$$\Pi_M(s) = 1 \quad \text{iff} \quad M(s) = 1 \wedge t_M(s) < t_B(s), \quad (6)$$

with separate fields for source grade, residual difference, and named priors. Priority on s is not priority on every neighboring idea or on the target's entire theory.

3.5 Category and process boundary

Both accounts can be mapped into a small causal category whose objects are local physical state, population wave, coordinated processing, and consciousness-relevant state. The corresponding functors preserve the order of that bounded chain. They do not preserve every internal object or morphism: NAPOT, PWD content, and the potassium / APD / calcium route have no target-side image. The result is a structure-preserving span through a shared abstraction, not a categorical isomorphism of complete theories.

At the process level, the shared observable trace is observe-local-state -> aggregate-wave -> coordinate-population -> affect-conscious-processing. Block's final transition remains explicitly speculative. SAN has additional visible actions, while Block has evolutionary and glial actions absent from the 2017 source. The audit therefore records bounded observational trace equivalence, not a checked pi-calculus bisimulation.

4. Atom-by-atom results

4.1 Broad atoms that do not yield a Micah-priority claim

The first boundary is anti-functionalism. Block's objections to functionalism date to 1978 [1]. SAN's insistence on physical, molecular, and oscillatory implementation overlaps this broad stance, but the chronology does not support claiming that Blumberg introduced the general roles-versus-realizers issue before Block.

The second boundary is oscillatory binding. Block discussed Crick and Koch's 35-75 Hz hypothesis in 1995 and questioned its explanatory force [2,7]. The underlying proposal belongs to Crick and Koch; the broader history of brain rhythms is older still. SAN does not claim to have originated the idea that oscillations may matter for neural coordination.

These concessions are atom-specific. They do not become a "Block floor" that automatically owns later, more specific combinations.

4.2 The bounded ionic-to-wave-to-consciousness match

The strongest comparison is the following:

Causal position	Blumberg, April 2017	Block, 2025-2026	Disposition
Input	time-varying spatial pattern of ionic charges in a neural matrix	extracellular ionic fluctuations, glial regulation, and electrochemical processing	partial-equivalent input class
Transform	brain-wave / electromagnetic communication, interaction, and feedback	population-wide electrical waves and brain rhythms that help drive neural firing and sophisticated processing	partial-equivalent transform
Output	conscious-versus-unconscious communication protocol and internal representation	speculative dependence of consciousness on electrochemically generated rhythms	bounded output convergence

This chain passes the mechanism lens because the same ordered causal behavior is retained at a useful level of abstraction. It passes the narrative lens because both accounts move from local ionic / electrochemical state to distributed wave organization to consciousness relevance. It passes Detective Mode because the 2026

conclusion can be traced backward through the 2025 candidate mechanism, while the 2017 source supplies an earlier public joined chain.

The result is a partial match, not a lower-quality nonmatch. “Partial” specifies scope: the overlapping causal subchain is real, while residual components remain author-specific.

4.3 SAN residuals that cannot be imported into Block

Block’s papers do not state that phase-wave differentials are the content variable of experience. They do not define dendritic arrays as receive-transform-project screens. They do not propose NAPOT, oscillatory tomography, a volumetric neural rendering, coincidence-as-bit, or the SAN potassium-to-action-potential-duration-to-calcium-to-release chain. These residuals matter in two ways.

First, they protect the distinctiveness of SAN. Second, they prevent an invalid translation in which the analyst silently substitutes SAN’s detailed mechanism for Block’s broader words and then claims full identity. The correct map is containment of a bounded subchain, not equivalence of all details.

4.4 Target-side residuals and named donors

Block emphasizes evolutionary comparisons among purely electrical and electrochemical nervous systems, the role of extracellular fluid and glia, and a substrate requirement for biological consciousness that may exclude some artificial systems [5,6]. His 2025 text attributes ionic-wave evidence to Fields and large-scale dynamic-pattern reasoning to Godfrey-Smith [5,9-11]. Those sources must remain visible in the genealogy.

The Fields paper was first published online on May 31, 2025. Its abstract proposes that field-potential oscillations and glia may help explain vertebrate cognitive power under weak cortical synaptic connectivity [11]. This is a direct target-side donor route. It does not erase Blumberg’s 2017 earlier joined ionic-pattern / brain-wave / conscious-feedback formulation [14]; it shows that Block’s later wording participates in an identifiable contemporary literature rather than originating every component itself.

5. Before / after result

The longitudinal record changes in a way that a canonical-paper-only comparison would miss.

Before SAN’s public 2017 mechanism, Block’s work already opposed reductive functional sufficiency and sought neural realizers. It also discussed oscillatory consciousness hypotheses through named prior literature [1-3,7]. But the reviewed 1978, 1995, and 2007 texts do not affirm an ionic-to-population-wave-to-consciousness causal mechanism [1-3]. The 2018 machine-consciousness letter still focuses on subjective experience and the insufficiency of global availability or metacognition, without the electrochemical-wave chain [4].

After the 2017 SAN source and the 2022 public Git maturation [14-16], Block’s 2025 paper introduces electrochemical computation, extracellular ionic fluctuations, brain rhythms, and large-scale dynamics as candidate consciousness-relevant biological realizers [5]. The 2026 follow-up then states the electrochemical-processing-to-brain-rhythm-to-consciousness sequence directly [6].

This satisfies the controlled longitudinal category LSM-COMPONENT-SHIFT: components that were absent from the reviewed earlier target stages become central in later stages and move toward a bounded SAN signature. The finding remains content- and chronology-specific. No evidence reviewed here shows that Block accessed SAN, and the paper makes no claim about his intent or source exposure.

6. What the comparison establishes

The audit yields five paper-local favorable dispositions for Blumberg:

1. earlier public articulation of a time-varying ionic neural pattern used as representational state;
2. earlier joining of that ionic state to brain-wave or electromagnetic communication;
3. earlier joining of the wave process to feedback-based conscious versus unconscious processing;
4. earlier publication of the combined ionic-state → wave-communication → consciousness-related subchain; and
5. earlier publication of a broader distributed field / rendering architecture that partially contains Block's later large-scale-dynamics narrative, while preserving Godfrey-Smith's donor status on Block's route.

The first four are mechanism-level bounded matches. The fifth is a narrative-level partial and is therefore reported separately. The paper does not claim Blumberg priority over anti-functionalism, general biological naturalism, general oscillatory neuroscience, gamma binding, electrochemical neurotransmission, glial physiology, or the phenomenal / access distinction.

7. Why no numerical derivation probability is reported

An earlier private audit assigned five strong and seven partial matches and calculated Bayes factors of 78,125 and approximately 55 under two assumed base-rate scenarios. Those calculations remain preserved as historical audit artifacts, but they are not adopted here.

The reasons are methodological. The atom set was selected after examining the target; several atoms are causally dependent; field-wide base rates were not measured; no control-author corpus was used; no access model was established; and no out-of-sample validation was performed. Multiplying analyst-chosen probabilities under those conditions produces a number, not a calibrated evidential probability.

The strongest defensible statement is qualitative: the dated record shows a nontrivial, bounded, later convergence on a joined causal chain that Blumberg publicly articulated earlier. The evidence does not currently permit a numerical estimate of independent versus dependent derivation.

8. Testable scientific differences

The comparison becomes more useful when it produces discriminating experiments rather than only chronology.

8.1 Block-style substrate test

Compare systems implementing similar computational roles with different physical realizers. If biological electrochemical dynamics are necessary, role-matched nonbiological systems should lack one or more consciousness-relevant markers. The difficulty is that no agreed independent consciousness measure currently resolves the test.

8.2 SAN-specific phase-content test

SAN predicts more than the importance of waves. It predicts that controlled, receiver-relative phase changes in an appropriate recurrent array should alter represented content in a structured way. A successful test would require decoding content-specific phase relations beyond changes in arousal, firing rate, spectral power, or generic synchrony.

8.3 Molecular-to-wave perturbation test

Manipulate potassium-current kinetics within a preparation while measuring action-potential duration, calcium entry, release statistics, local phase relations, and downstream representational decoding. The SAN chain is supported only if the multiscale effects follow the predicted ordered mediation and outperform alternative explanations based on rate, excitability, or network-state change alone.

8.4 Extracellular / glial wave test

Perturb extracellular ion regulation or glial contributions while controlling synaptic and intrinsic excitability. Block's newer proposal gains support if the manipulation selectively changes brain-rhythm organization and sophisticated processing in the predicted direction. A result that changes excitability without a specific rhythm-mediated functional effect would be insufficient.

9. Limitations

This is a source-based conceptual and chronological comparison, not an experimental adjudication of either theory. The 2026 Block follow-up is listed by the author as forthcoming and may change before final publication. The SAN genealogy contains both early conceptual ancestors and later mature operators; this paper does not backdate 2022 terms into 2011, 2012, or 2017. Some early public-source timestamps have different evidential grades, which should remain visible in any final exhibit packet.

The comparison also does not determine the sociological route by which similar ideas arise. Explicit citations to Fields, Godfrey-Smith, Crick, Koch, and other prior work establish part of Block's intellectual context, but absence of a SAN citation does not by itself establish awareness or intent.

10. Conclusion

Block's 2025-2026 work should not be described as the origin of the entire role / realizer, electrochemical, brain-rhythm, or consciousness landscape. His earlier anti-functionalism is genuine; his later mechanism also relies on named contemporary and historical donors. At the same time, the absence of Block's full adoption of NAPOT or PWD does not erase the narrower causal match.

The primary sources support a precise priority result: Micah Blumberg publicly joined a changing ionic neural state, brain-wave / electromagnetic communication, feedback, and consciousness-related processing in April 2017 [14]. Block's 2025-2026 work later moves into a partially equivalent chain linking electrochemical or ionic dynamics to brain rhythms, sophisticated processing, and possible consciousness necessity [5,6]. Blumberg's 2022 and 2024 work then shows that the earlier 2017 chain was not an isolated metaphor but an ancestor of a more detailed SAN architecture [15-17].

That is the defensible finding: a bounded Micah-earlier mechanism and narrative priority, preserved alongside real prior art, target-specific residuals, and an explicit refusal to convert uncalibrated similarity counts into claims about access or intent.

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